

Xinhao Liu

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Research Interests

My research interests lie at the intersection of robotics, computer vision, and deep learning, with a focus on embodied AI in open-world environments. My research focuses on *Universal Navigation*, where I aim to build embodied intelligence that can go anywhere that humans can go.

Keywords: World Modeling, Video Understanding, Digital Twins

Education

2023 – present **Ph.D. in Computer Science**, New York University, New York, NY.

Advisor: [Prof. Chen Feng](#)

2019 – 2023 **B.Sc. in Mathematics**, New York University, Shanghai, China.

Major: Honors Mathematics | Data Science

Experience

Nov 2025 – present **MTS Intern**, Amazon FAR, San Francisco, CA

May – Aug 2025 **Research Intern**, Waymo, Mountain View, CA

2021 – 2023 **Research Intern**, AI4CE Lab, New York University, New York, NY

Jun – Aug 2021 **Intern Developer**, FNZ, Shanghai, China

Research Publications

First-authored papers (including equal contribution): 5 Citations: 316 H-index: 6

Full publication list on [Google Scholar](#).

* indicates equal contribution.

Conference Proceedings

- 0 **Xinhao Liu**, Jiaqi Li, Youming Deng, Ruxin Chen, Yingjia Zhang, Yifei Ma, Li Guo, Yiming Li, Jing Zhang, and Chen Feng, “Wanderland: Geometrically grounded simulation for open-world embodied ai,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 1041–1052.
- 1 **Xinhao Liu***, Jintong Li*, *et al.*, “Citywalker: Learning embodied urban navigation from web-scale videos,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 6875–6885.
- 2 Yiming Li*, Sihang Li*, **Xinhao Liu***, Moonjun Gong*, Kenan Li, Nuo Chen, Zijun Wang, Zhiheng Li, Tao Jiang, Fisher Yu, Yue Wang, Hang Zhao, Zhiding Yu, and Chen Feng, “Sscbench: Monocular 3d semantic scene completion benchmark in street views,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2024.
- 3 **Xinhao Liu***, Moonjun Gong*, Qi Fang, Haoyu Xie, Yiming Li, Hang Zhao, and Chen Feng, “Lidar-based 4d occupancy completion and forecasting,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2024.

- 4 Juexiao Zhang, Gao Zhu, Sihang Li, **Xinhao Liu**, Haorui Song, Xinran Tang, and Chen Feng, “Multiview scene graph,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- 5 Chao Chen*, **Xinhao Liu***, Yiming Li, Li Ding, and Chen Feng, “Deepmapping2: Self-supervised large-scale lidar map optimization,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 9306–9316.

Journal Articles

- 0 Chao Chen, Zegang Cheng, **Xinhao Liu**, Yiming Li, Li Ding, Ruoyu Wang, and Chen Feng, “Self-supervised place recognition by refining temporal and featural pseudo labels from panoramic data,” *IEEE Robotics and Automation Letters*, vol. 10, no. 1, pp. 248–255, 2025. doi: [10.1109/LRA.2024.3495584](https://doi.org/10.1109/LRA.2024.3495584).

Honors & Awards

- 2025 **Deborah Rosenthal, MD Award**, New York University, New York, NY. Awarded for outstanding performance on PhD qualifying exam (2 awardees per year).
- 2024 **Conference Attendance Grant**, New York University, New York, NY. (\$1,500 grant)
- 2023 **University Honors Scholar**, New York University, New York, NY.
- 2023 **Student Travel Grant for Conference**, NYU Shanghai, Shanghai, China. (\$4,000 grant)
- 2023 **Dean’s List of Academic Year**, NYU Shanghai, Shanghai, China.
- 2022 **Dean’s List of Academic Year**, NYU Shanghai, Shanghai, China.
- 2022 **UGSRP Research Fund**, NYU Tandon, New York, NY. (\$5,000 fund)
- 2020 **Best Research Project**, Undergraduate Research Symposium, NYU Shanghai, Shanghai, China. (2 out of 30+ projects)
- 2020 **Dean’s Undergraduate Research Fund**, NYU Shanghai, Shanghai, China. (¥7,000 fund)

Invited Talks

- 2025 **Amazon FAR**, *CityWalker: Learning Embodied Urban Navigation from Web-Scale Videos*.

Grants

- 2025 **The START Program**, Samsung Research America. *Action-Conditioned Video Generation for Dynamic Environments* (Rejected). **Role**: Project Collaborator, proposal writing, leading the project.
- 2024 **The NVIDIA Academic Grant Program**, NVIDIA. *Open Digital Twins for New York City*. **Amount**: 8 NVIDIA RTX 6000 Ada Graphics Cards (worth \$54,400). **Role**: Project Collaborator, main proposal writing and project lead.

Professional Services

- Reviewer** **NeurIPS** (2025), **ICCV**, (2025), **CVPR** (2024–2026), **ICLR** (2024–2026), **IROS** (2024–2025), **ICRA**, (2023).